Debian based Computer Literacy
Setup and Everyday Reference

Hugh Rodgers

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1 Install VS Code or VSCodium

Choose one editor. For VS Code, download the Debian package from the official Linux setup guide and install it with the graphical software centre or the command shown there or run sudo apt install ./vscode.deb. For VSCodium, follow theofficial installation options; choose the package source for your distribution. On a managed computer, follow university policy before installing software.

2 Install the shared tools

On Debian or Ubuntu, update the package list and install Python, its virtual environment and package tools, and Git:

```
sudo apt update
sudo apt install git python3 python3-venv python3-pip
```

For a local LaTeX installation, add the compiler and workshop packages:

```
sudo apt install latexmk texlive-latex-recommended texlive-latex-extra texlive-
    pictures
```

Review package-manager changes before confirming. Do not make system changes on a managed university computer without permission. These package names and commands are for Debian-based systems; other distributions use different package managers. See the Debian package-management documentation, Python downloads, Git for Linux, and TeX Live documentation. Overleaf is a browser-based alternative to installing TeX locally.

```
python3 --version
git --version
pwd
```

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3 Terminal, projects, and paths

The terminal runs a shell, which accepts commands to navigate folders and start programs. A project is an organised directory; relative paths begin from the current working directory. Linux filenames are case-sensitive, and hidden names begin with a dot, such as `.venv` and `.git`. Quote paths that contain spaces.

```
pendulum-lab/
  README.md
  requirements.txt
  data/
    raw_measurements.csv
  analysis/
    analysis.py
    analysis.ipynb
  figures/
  report/
    report.tex
```

Task	Bash command
Current directory	`pwd`
List files, including hidden	`ls -la`
Move to the parent directory	`cd ..`
Enter a path with spaces	`cd "My Project"`
Copy / move a file	`cp source.csv copy.csv` / `mv copy.csv data.csv`
Find CSV files	`find . -name '*.csv'`
Search file text	`grep -n 'time' notes.txt`
Open the current folder	`xdg-open .`, on a desktop system
Open the folder in the editor	`code .` or `codium .`

Inspect paths before copying or moving files: an existing destination may be overwritten. Prefer the file manager's Trash when removing files. Command behaviour is documented in the GNU Coreutils manual and the GNU Findutils manual.

4 Python and project environments

Create one virtual environment in the project directory containing `requirements.txt`. Install project packages inside it, not into the operating system's Python:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python -m pip check
python -c "import sys; print(sys.executable)"
deactivate
```

The printed interpreter path should include `.venv`. A virtual environment keeps project packages separate; it is not a security sandbox, and it should not be committed or copied to another computer. See the Python virtual-environment documentation. If creating one reports a missing `ensurepip`, install the matching distribution package with administrator permission. Do not use `sudo pip` or bypass the distribution's externally-managed Python protection.

5 Git and LaTeX

Git works locally; GitHub hosts online copies and collaboration. Saving a file is not the same as committing it, and a commit is not the same as pushing it. For the shared commands and safe workflow, see the Git and GitHub guide.

LaTeX source (`.tex`) is compiled into a PDF. Use Overleaf in a browser or compile locally with the TeX Live installation above. From the repository root, build this handout with:

```
latexmk -cd -pdf -interaction=nonstopmode -halt-on-error Linux/Debian_Bassed/Linux.TEX
```

The shared LaTeX guide covers report structure, equations, figures, and references; see also the Overleaf LaTeX introduction and LaTeX Project documentation.

6 Shortcuts and Bash configuration

- `Ctrl+C` interrupts the running command; `Ctrl+D` ends input or exits an empty shell.
- `Tab` completes paths; `Up` recalls command history; `Ctrl+R` searches Bash history.
- `Ctrl+Shift+C/V` copies and pastes in many Linux terminal applications. Desktop shortcuts can vary.

Bash reads `~/.bashrc` for interactive non-login terminals. To add personal shortcuts, open that file in a text editor and append simple aliases such as:

```
alias ll='ls -lah'
alias croot='cd ~'
```

Save the file, then load the changes in the current terminal:

```
source ~/.bashrc
```

Back up and review your own file before editing it. Never replace it wholesale with another person's configuration. See the Bash startup-file documentation.

7 Troubleshooting

Command not found: Check that the tool is installed and reopen the terminal after installation.

Permission required: Check university policy; do not use administrator privileges as a general fix.

Python package installation is blocked: Create and activate the project's virtual environment.

File not found: Check `pwd`, filename case, and the relative path.

No graphical window over SSH: The remote machine may not have a display.

LaTeX package missing: Use the TeX distribution's package manager or Overleaf; an editor extension is not a TeX distribution.

For Python, fitting, and data-analysis exercises, use the shared exercise notebook.